2022 Minnesota Academic Standards in Mathematics Learning Progressions – Strand 2 Spatial Reasoning

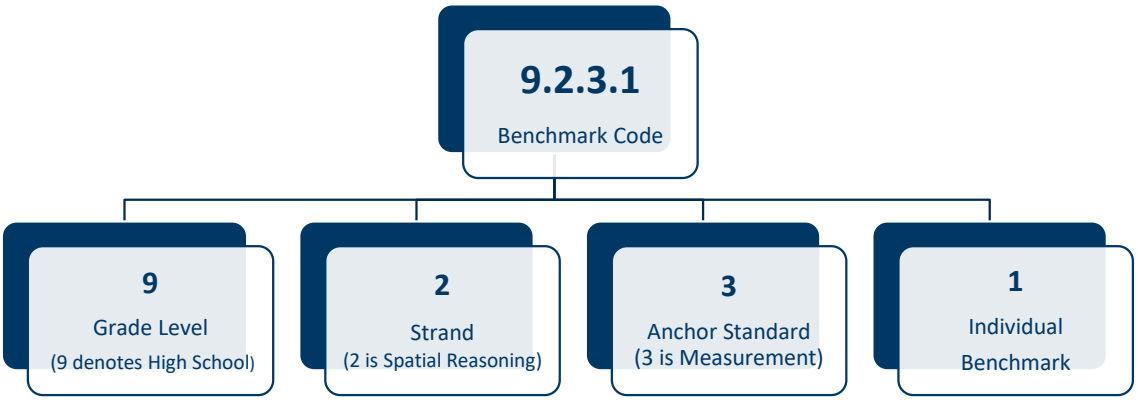
The 2022 Minnesota Academic Standards in Mathematics in the vertical format show the standards, organized by strand, with their corresponding grade-level benchmarks for grades K–High School. These learning progressions allow teachers, school leaders and administrators, and parents and caregivers to see what students should be learning in their grade level in relation to the grade levels above and below.

The 2022 Minnesota Academic Standards in Mathematics are organized into three strands: 1) Data & Probability; 2) Spatial Reasoning; and 3) Patterns & Relationships. Each strand has a series of two to three anchor standards and each anchor standard has one or more grade-level benchmarks with a 4-digit code (e.g., 0.1.1.1).

This document shows the learning progression of the Data and Probability strand from kindergarten through 12th grade, organized by anchor standards and individual benchmarks. Importantly, the standard and benchmark information below is the same as can be found in the primary Minnesota Academic Standards in Mathematics documents—available on the MDE Math implementation webpage—but is organized in a learning progression format. Please see the Spatial Reasoning and Pattern & Relationships learning progression documents, also available on the MDE Math implementation webpage.

Reading the Mathematics Standards and Benchmarks

Each benchmark has a four-digit code, which includes the grade level, strand, standard and benchmark. Each benchmark is followed by two dimensions; Mathematical Practices (MP) and Benchmark Contexts.



Dimension 1: Mathematical Practices (MP)
• MP1: Make sense of problems and persevere in solving them. • MP2: Reason abstractly and quantitatively. • MP3: Construct viable arguments and critique the reasoning of others. • MP4: Model with mathematics. • MP5: Use appropriate tools strategically. • MP6: Attend to precision. • MP7: Look for and make use of structure. • MP8: Look for and express regularity in repeated reasoning.

Dimension 2: Benchmark Contexts
• ✣ Minnesota Tribal Nations Contexts • $ Financial Literacy Contexts • # Computer Science Contexts • µ Modeling Contexts • ☼ Real-world Contexts

Strand 2: Spatial Reasoning

Students will actively make sense of relationships between and within geometric figures, generalize statements about the figures and develop arguments about these relationships.

Anchor Standard 3

Measurement

Investigate measurement using a variety of tools, units, systems, processes and techniques in various cultures. Explain and reason with attributes, estimations and formulas to communicate measurement(s) and relationships effectively. Justify decisions and consider the reasonableness of the measurement.

A. Attribute Measurement

B. Area and Volume

Anchor Standard 4

Geometry

Analyze characteristics of geometric shapes to make mathematical arguments and justifications about geometric relationships. Use visualization and geometric modeling to compare, solve problems and communicate ideas.

A. Shape Properties, Classification and Logical Argumentation

B. Spatial Reasoning, Transformations and Geometric Modeling

Anchor Standard Three: Measurement

Investigate measurement using a variety of tools, units, systems, processes and techniques in various cultures. Explain and reason with attributes, estimations and formulas to communicate measurement(s) and relationships effectively. Justify decisions and consider the reasonableness of the measurement.

3A. Attribute Measurement

Grade	Code	Benchmark
K▼	0.2.3.1	Compare objects with a measurable attribute in common, to which object has "more of," "less of" or the "same as" the attribute and explain the reasoning. (MP3, MP5) ✣ $ ☼
	0.2.3.2	Describe several measurable attributes of objects such as length and weight. (MP4, MP6) ✣ ☼
1	1.2.3.1	Order three objects by length. Compare the lengths of two objects indirectly by using a third object. (MP2, MP5) ✣ ☼
	1.2.3.2	Measure the length of an object in terms of non-standard units. (MP5) ✣ ☼
	1.2.3.3	Identify pennies, nickels and dimes. Find the value of a group of these coins, up to one dollar. (MP1, MP7) $
2	2.2.3.1	Estimate lengths using units of inches, feet, centimeters and meters. (MP2, MP5) ✣ ☼
	2.2.3.2	Measure to determine how much longer one object is than another, expressing the length difference in terms of a standard-length unit. Relate addition and subtraction to length. (MP5) ☼
	2.2.3.3	Measure the length of an object by selecting and using appropriate tools such as rulers, yardsticks, meter sticks and measuring tapes. (MP5, MP6) ✣ ☼
	2.2.3.4	Represent whole numbers as lengths from 0 on a number line with equally spaced points corresponding to the numbers 0, 1, 2, … Represent whole-number sums and differences, within 100, on a number line. (MP4, MP5)
	2.2.3.6	Use addition and subtraction, within 100, to solve contextual situations involving lengths that are given in the same units using drawings (such as rulers) and equations with a symbol for the unknown number to represent the situation. (MP1, MP4) $
	2.2.3.7	Identify pennies, nickels, dimes and quarters. Find the value of a group of coins and determine combinations of coins that equal a given amount, using $ and ¢ symbols appropriately. (MP7) $
3	3.2.3.1	Measure lengths to the nearest fourth when measuring with standard units. (MP2, MP5, MP6) ☼
	3.2.3.2	Compare and contrast the relative sizes of measurement units within one system (such as inches, feet, and yards; centimeters and meters; grams and kilograms; ounces and pounds; cups and gallons; quarters and dollars). (MP5, MP6) $ ☼
	3.2.3.3	Calculate the perimeter of a polygon with whole number side lengths. (MP5, MP6)

Grade	Code	Benchmark
3	3.2.3.4	Use addition and subtraction with whole numbers, within 100, to calculate change up to one dollar in several different ways, using $ and ¢ symbols appropriately. (MP7, MP8) $
4	4.2.3.1	Classify angles as acute, right and obtuse by estimation, comparison with a right angle and by measurement. (MP6) ✣
	4.2.3.2	Measure angles with a protractor. (MP5) ☼
	4.2.3.3	Determine the perimeter and area of two-dimensional figures and label with appropriate units. (MP2, MP6) ✣ ☼
	4.2.3.7	Make change up to $20 with place values, using $ and ¢ symbols appropriately. (MP7) $
5	No Code	No benchmark at this level.
6	6.2.3.3	Solve situations in various contexts involving conversion of time, weights, capacities, lengths and area within measurement systems using appropriate units. (MP4, MP5) ✣ ☼
	6.2.3.4	Estimate time, weights, capacities, lengths and dollar amounts using benchmarks in measurement systems with appropriate units. (MP1) ✣ $ ☼
7	7.2.3.1	Demonstrate an understanding of the proportional relationship between the diameter and circumference of a circle and that the unit rate (constant of proportionality) is π. (MP1, MP2) ✣ ☼
	7.2.3.2	Calculate the circumference and area of circles to solve situations in various contexts. (MP4, MP6) ✣ μ ☼
	7.2.3.3	Calculate the arc length and area of sectors of circles (given the central angle) to solve situations in various contexts. (MP4, MP7) ✣ ☼
8	8.2.3.2	Apply the Pythagorean Theorem to solve multi-step contextual situations by determining unknown side lengths in right triangles in both two- and three-dimensional shapes. (MP4) μ ☼
	8.2.3.3	Determine the distance between two points on a horizontal or vertical line in a coordinate system. Use the Pythagorean Theorem to find the distance between any two points in a coordinate system. (MP7) ✣ ☼
▲High School	9.2.3.1	Apply the Pythagorean Theorem and distance formula to compute perimeters of polygons and areas of right triangles and rectangles on coordinate planes. (MP1) ☼
	9.2.3.2	Apply properties of 45°-45°-90° and 30°-60°-90° triangles to solve situations and logically justify results. (MP1, MP4) ☼
	9.2.3.6	Use units of measure and dimensional analysis to solve multi-step situations. Interpret units consistently in formulas. Interpret the scale and the origin in graphs and data displays. (MP1, MP4) ✣ $ ☼
	9.2.3.8	Use similarity to determine the side ratios in right triangles as properties of the angles in the triangle, leading to definitions of trigonometric ratios for acute angles. (MP8)

Anchor Standard Three: Measurement

Investigate measurement using a variety of tools, units, systems, processes and techniques in various cultures. Explain and reason with attributes, estimations and formulas to communicate measurement(s) and relationships effectively. Justify decisions and consider the reasonableness of the measurement.

3B. Area and Volume Measurement

Grade	Code	Benchmark
K▼		No benchmark at this level.
1		No benchmark at this level.
2	2.2.3.5	Partition a rectangle into rows and columns of same-size squares and count the total number. (MP8)
3	3.2.3.3	Calculate the perimeter of a polygon with whole number side lengths. (MP5, MP6)
4	4.2.3.4	Determine the perimeter and area of two-dimensional figures and label with appropriate units. (MP2, MP6) ✣ ☼
	4.2.3.5	Find the areas of geometric figures that can be decomposed into rectangular shapes using tools like dot or grid paper. Label area measurements using square units. (MP2)
	4.2.3.6	Explain why the area of a rectangle can be calculated by multiplying the length by the width and use the formula $A = l \text{ x } w$ to calculate the area of rectangles with whole number side lengths. (MP3)
5	5.2.3.1	Develop, justify and use formulas to determine the area of parallelograms and triangles. Find the areas of polygons that can be decomposed into parallelograms and triangles. (MP3) # µ
	5.2.3.2	Estimate the area of two-dimensional shapes, both polygons and non-polygons, using tools such as dot or grid paper. (MP1, MP5)
	5.2.3.3	Use unit cubes to measure volume. Describe a unit cube as a cube with side length 1 unit that is said to have "one cubic unit" of volume and can be used to measure volume. (MP5, MP6)
	5.2.3.4	Use various strategies to measure the volume and surface area of three-dimensional shapes made of a collection of unit cubes. (MP5, MP6) ✣ ☼
	5.2.3.5	Find the volume of a right rectangular prism with whole-number side lengths by packing it with unit cubes. Show that the volume is the same by unit cubes as by multiplying the edge lengths (l x w x h) or by multiplying the height by the area of the base. (MP3, MP8) ☼
6	6.2.3.1	Calculate the measurements of the surface area of rectangular and triangular prisms using appropriate units. Justify the formulas used. Justification may involve decomposition, nets or other models. (MP1, MP3) ✣ # ☼
	6.2.3.2	Calculate the measurement of the volume of prisms (including triangular and nonrectangular prisms) using appropriate units. Justify the formulas used. Justification may involve decomposition or other models. (MP1, MP7) ✣ # ☼

Grade	Code	Benchmark
6	6.2.3.5	Find the area of special quadrilaterals and polygons by composing into rectangles or decomposing into triangles and other shapes. Apply these techniques in contextual mathematical situations. (MP2, MP5) ☼
7	7.2.3.2	Calculate the circumference and area of circles to solve situations in various contexts. (MP4, MP6) ✜ µ ☼
	7.2.3.3	Calculate the arc length and area of sectors of circles (given the central angle) to solve situations in various contexts. (MP4, MP7) ✜ ☼
	7.2.3.4	Calculate the surface area and volume of cylinders. Justify the formulas used. (MP3, MP8) ✜ ☼
8	8.2.3.1	Informally justify the Pythagorean Theorem and its converse by using measurements, diagrams or computer software. (MP2, MP3) ✜ ☼
▲High School	9.2.3.1	Apply the Pythagorean Theorem and distance formula to compute perimeters of polygons and areas of right triangles and rectangles on coordinate planes. (MP1) ☼
	9.2.3.3	Use trigonometric ratios and the Pythagorean Theorem and its converse to solve contextual situations, including right triangles or polygons that can be decomposed into triangles. (MP4, MP7) ☼
	9.2.3.4	Develop, justify and use decomposition to determine the formula for surface area and volume of various three-dimensional figures. (MP7, MP8) ✜ # ☼
	9.2.3.5	Use the formulas for surface area and volume of various three-dimensional figures to solve multi-step contextual situations. (MP4, MP5) ✜ ☼
	9.2.3.7	Compute the unit cost and total cost to cover different areas and volumes in applications, such as painting, carpeting and gardening. Optimize costs for various models, including linear and quadratic. Use technology, including spreadsheets. (MP4, MP5) $ µ ☼
	9.2.3.9	Apply dilations of scale factor k on length, area and volume and recognize the effect is multiplication by k, k^2, k^3 respectively. (MP2) µ ☼

Anchor Standard Four: Geometry

Analyze characteristics of geometric shapes to make mathematical arguments and justifications about geometric relationships. Use visualization and geometric modeling to compare, solve problems and communicate ideas.

A. Shape Properties, Classification, and Logical Argumentation

Grade	Code	Benchmark
K▼	0.2.4.1	Sort objects using characteristics such as shape, size, color and thickness. (MP1, MP3) ✣ ☼
	0.2.4.2	Identify and compare two- and three-dimensional shapes such as squares, circles, triangles, rectangles, trapezoids, hexagons, cubes, cones, cylinders and spheres using informal language to describe their similarities, differences, parts and other attributes. (MP2) ✣ ☼
	0.2.4.3	Compose, decompose and name simple shapes. Recognize shapes regardless of their overall size and orientation. (MP1, MP2) µ
1	1.2.4.1	Describe attributes of two- and three-dimensional objects, such as triangles, squares, rectangles, circles, rectangular prisms, cylinders, cones and spheres. (MP7, MP8) ✣ ☼
	1.2.4.2	Compose two-dimensional shapes (rectangles, squares, trapezoids, triangles, half-circles and quarter-circles) to create a composite shape. Decompose composite shapes into triangles, rectangles, squares and sectors. (MP3, MP7) µ
	1.2.4.4	Identify shapes regardless of their orientations. (MP1, MP6) ☼
2	2.2.4.1	Classify two- and three-dimensional figures according to the number and shape of faces and the number of sides, edges and vertices. (MP1, MP2) ✣ ☼
	2.2.4.2	Create a representation for basic two-dimensional shapes such as squares, circles, triangles, rectangles, trapezoids and hexagons. (MP5, MP6) ☼
3	3.2.4.1	Create representations of regular and irregular polygons with a given number of sides, including triangles, quadrilaterals, pentagons, hexagons and octagons. (MP2, MP6) ✣ ☼
4	4.2.4.2	Create representations of triangles given the relationships among the sides (scalene, isosceles, equilateral) and the angles (acute, right, obtuse). (MP1, MP6) ☼
	4.2.4.3	Sort and classify quadrilaterals in a hierarchy, including squares, rectangles, trapezoids, rhombuses, parallelograms and kites. Understand that attributes belonging to a category of two-dimensional figures also belong to all subcategories of that category. (MP1, MP6) ✣ µ ☼
	4.2.4.4	Create a representation and describe the front, top and side views of three-dimensional figures composed of cubes and rectangular prisms. (MP5)
	4.2.4.5	Draw the nets of cubes. Recognize nets that will and will not form cubes. (MP1)
5	5.2.4.1	Classify and describe prisms and pyramids by their defining attributes and the number of edges, faces, vertices and bases. (MP1, MP6) ✣ ☼
	5.2.4.2	Recognize, draw and compare different nets for prisms, pyramids, cylinders and cones. (MP1)
6	6.2.4.2	Decompose polygons into triangles to investigate the sum of the interior angles of polygons. (MP1, MP7) #
7		No benchmark at this level.
8		No benchmark at this level.
▲High School	9.2.4.2	Apply properties of equilateral, isosceles and scalene triangles to solve situations and logically justify results. (MP1, MP3) ✣ ☼

Grade	Code	Benchmark
▲**High School**	9.2.4.3	Apply properties such as interior and exterior angles and parallel and perpendicular sides to define, classify and solve situations involving regular and irregular polygons. (MP1) ✣ ☼
	9.2.4.5	Accurately interpret and use words and phrases such as “if ... then,” “if and only if,” “all” and “not.” Recognize the logical relationships between an “if ... then” statement and its inverse, converse and contrapositive. (MP6, MP7) ✣ # ☼
	9.2.4.6	Analyze the validity of a logical argument and give counterexamples to disprove a statement. (MP1, MP3) ✣ # ☼
	9.2.4.7	Construct logical arguments using axioms, definitions, theorems and postulates that clearly justify the reasoning. (MP3, MP6) #
	9.2.4.8	Identify, describe and solve situations using relationships among inscribed angles, circumscribed angles, radii and chords in circles. (MP5) ✣ ☼
	9.2.4.10	Apply properties of similar figures to solve situations and logically justify results. (MP3) ✣
	9.2.4.11	Show that two triangles are similar using the AA, SAS and SSS similarity criteria. (MP7)

Anchor Standard Four: Geometry

Analyze characteristics of geometric shapes to make mathematical arguments and justifications about geometric relationships. Use visualization and geometric modeling to compare, solve problems and communicate ideas.

B. Spatial Reasoning, Transformations, and Geometric Modeling

Grade	Code	Benchmark
K▼	0.2.4.4	Describe objects in the environment using names of shapes. Describe the relative positions of these objects using terms such as above, below, beside, in front of, behind and next to. (MP1, MP6) ✣ ☼
1	1.2.4.3	Describe objects in the environment using names of shapes and describe the relative positions of these objects using left and right. (MP6) ☼
2	2.2.4.3	Describe the location of an object in relation to another object. (MP1, MP6) ☼
3	3.2.4.1	Create representations of regular and irregular polygons with a given number of sides, including triangles, quadrilaterals, pentagons, hexagons and octagons. (MP2, MP6) ✣ ☼
4	4.2.4.1	Draw points, line segments, rays, lines, angles and perpendicular and parallel lines. Identify these in two-dimensional figures. (MP5) ☼
5	5.2.4.2	Recognize, draw and compare different nets for prisms, pyramids, cylinders and cones. (MP1)
6	6.2.4.1	Determine missing angle measures in a triangle using the fact that the sum of the interior angles of a triangle is 180°. Use models to illustrate this fact. (MP8, MP2) ✣ ☼
	6.2.4.3	Draw polygons in the coordinate plane given coordinates for the vertices. Use coordinates to find the length of a side joining points with the same first coordinate or the same second coordinate. Apply these techniques in contextual situations. (MP4, MP7) ☼
7	7.2.4.1	Graph and describe translations and reflections of figures on a coordinate grid. Determine the coordinates of the vertices of the figure after the transformation. Describe the properties of congruency when performing translations and reflections. (MP8)
	7.2.4.2	Describe the properties of similarity, distinguishing between similarity and congruency, compare geometric figures for similarity and determine scale factors. (MP1, MP6) ✣ ☼
	7.2.4.3	Apply scale factors, length ratios and area ratios to determine side lengths and areas of similar geometric figures. (MP5)
	7.2.4.4	Use proportional reasoning and ratios to solve situations involving scale drawings and conversions of measurement units. (MP2, MP4) ✣ $ ☼
8	8.2.4.1	Use similar triangles to explain why the slope m is the same between any two distinct points on a non-vertical line in the coordinate plane. Derive the equation $y = mx$ for a line through the origin and the equation $y = mx + b$ for a line intercepting the vertical axis at b. (MP8)
	8.2.4.2	Given a line on a coordinate plane and the coordinates of a point not on the line, find lines through that point that are parallel and perpendicular to the given line, using graphing technology or hand drawn graphs. (MP7) ✣
	8.2.4.3	Identify the different types of solutions possible for a system of linear equations (no solution, one solution, infinitely many solutions). Using slope, compare the number of solutions to the graphical representation of pairs of lines that are intersecting, parallel or identical. (MP2) $
▲High School	9.2.4.1	Apply properties of angles, including corresponding, exterior, interior, vertical, complementary and supplementary angles, to solve situations and logically justify results. (MP1, MP3)

Grade	Code	Benchmark
▲**High School**	9.2.4.4	Apply properties of congruent figures to solve situations and logically justify results. (MP1, MP3) ✣ ☼
	9.2.4.9	Describe and solve situations using the relationships of angles formed outside the circle for both secants and tangent lines. (MP7) ☼
	9.2.4.12	Make geometric constructions with a variety of tools, including dynamic geometric technology, to examine theorems, make and test conjectures, represent transformations and develop mathematical reasoning skills in multi-step situations. (MP5) ☼
	9.2.4.13	Create and compare rigid and non-rigid transformations and connect the preimage and image to congruence. (MP2)
	9.2.4.14	Describe the steps used to show various sequences of transformations, applying reflections, rotations, translations and/or dilations of geometric figures that map one figure onto the other. (MP8) ✣ #
	9.2.4.15	Apply geometric methods to solve design situations, including designing an object or structure to satisfy physical constraints or optimize resources. (MP4) $ µ
	9.2.4.16	Apply concepts of density based on area and volume in modeling contextual situations, including population density, BTUs per cubic foot and data storage. (MP4) ☼